# Mimaric

Cloud-Native Platform for Saudi Real Estate Lifecycle Management

## Technical White Paper

ميماريك :منصة سحابية لإدارة دورة حياة العقارات السعودية

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# PART I

## Context & Vision

## 1. Executive Summary

Mimaric is a Saudi-first PropTech SaaS platform that unifies the entire real estate lifecycle — from land acquisition and urban planning through construction, sales, leasing, maintenance, and financial reporting — under a single Arabic-first, bilingual interface. Built for Saudi real estate developers, property management companies, and facility managers, Mimaric replaces fragmented spreadsheets, WhatsApp coordination, and paper-based government compliance with an integrated digital platform.

### Key Differentiators

**Full Lifecycle Coverage:** 16 integrated modules spanning land acquisition to ongoing maintenance, eliminating the need for multiple disconnected tools.

**Saudi Regulation Native:** Schema-level alignment with 10 government platforms including Absher, MOC, Balady, Ejar, Wafi, ZATCA, PDPL, and NCA. Compliance is built into data models, not bolted on as an afterthought.

**Arabic-First Bilingual:** RTL-default interface with Hijri/Gregorian dual dates, SAR currency formatting with Riyal symbol, and IBM Plex Arabic typography. English is fully supported as a secondary language.

**Enterprise Security:** AES-256-GCM PII encryption at rest, 40+ granular RBAC permissions, SHA-256 search hashes, server-side data masking, and PDPL Article 32 audit trails with 8 action types.

**GIS-Integrated Planning:** Spatial subdivision planning with Leaflet mapping, Turf.js analysis, 8 layer types, compliance rule checking, and feasibility financial modeling.

**Cloud-Native Architecture:** Next.js 16 with Server Actions, React 19, Prisma 7 ORM, PostgreSQL on Supabase, and Turborepo monorepo for shared packages and incremental builds.

Target market: Saudi Arabia's $1.3 trillion real estate sector, currently undergoing Vision 2030 digital transformation with mandatory government platform adoption across rentals (Ejar), off-plan sales (Wafi), municipal services (Balady), tax compliance (ZATCA), and identity verification (Absher).

# 2. Market Context & Opportunity

## 2.1 Saudi Vision 2030 Real Estate Transformation

Saudi Arabia's Vision 2030 is driving unprecedented transformation in the real estate sector. The kingdom's ambitious giga-projects — NEOM, The Line, Red Sea Global, Diriyah Gate, and Jeddah Tower — combined with regulatory modernization through REGA (Real Estate General Authority), are creating massive demand for digital property management infrastructure.

**Key Market Dynamics**

- Housing demand of 1.2 million new units by 2030 to support population growth and urbanization
- REGA's mandate for digital contract registration: Ejar for rentals, Wafi for off-plan sales
- ZATCA's e-invoicing mandate (Phase 2) for all commercial transactions
- PDPL (Personal Data Protection Law) enforcement requiring data protection by design
- Municipal digitization through Balady platform for permits and licensing
- National identity verification ecosystem: Absher, Nafath, Yaqeen, Muqeem

## 2.2 The Problem: Operational Fragmentation

Saudi real estate companies today manage operations across disconnected systems:

- Spreadsheets for unit tracking, pricing matrices, and occupancy monitoring
- WhatsApp groups for maintenance coordination and tenant communication
- Paper forms for government compliance, contract execution, and audit trails
- Multiple government portals (Ejar, Balady, ZATCA, Absher) with manual, duplicated data entry
- Manual financial reconciliation across separate accounting, leasing, and maintenance systems

This fragmentation causes data loss, compliance gaps, delayed decision-making, poor tenant and buyer experience, and inability to scale operations beyond a handful of properties.

## 2.3 Market Size

| Metric | Value |
|---|---|
| Saudi real estate market (2025) | $1.3 trillion |
| PropTech adoption in GCC (CAGR) | 25% annual growth |
| Property management software (MENA, 2028) | $2.1 billion |
| New housing units needed by 2030 | 1.2 million |
| Arabic-first, regulation-native platforms | None dominant |

# 3. Problem Statement

Five core pain points define the gap Mimaric addresses:

**1. Data Silos —** Land acquisition, sales pipeline, leasing operations, maintenance work orders, and financial reporting exist in separate systems with no single source of truth. A change in unit status requires manual updates across multiple spreadsheets and portals.

**2. Manual Government Compliance —** Ejar contract registration, Balady building permits, ZATCA VAT invoicing, and PDPL audit trails all require manual data entry across different government portals. Each portal has its own login, data format, and submission workflow.

**3. No Arabic-First Enterprise Platform —** Existing property management solutions (Yardi, RealPage, AppFolio, Buildium) are Western-centric with English-first interfaces. Arabic support, when available, is a translation layer rather than a native implementation. RTL layout, Hijri dates, and SAR currency are afterthoughts.

**4. Lifecycle Gaps —** No single platform covers the full spectrum from raw land acquisition and urban subdivision planning through construction management, off-plan sales with escrow, lease management, and ongoing facility maintenance. Companies must integrate 4–6 separate tools.

**5. Security & Privacy Gaps —** Most property management solutions lack PDPL-compliant PII encryption at rest, NCA-aligned authentication policies, granular audit trails for data access, or proper role-based PII segregation. National IDs and financial data are often stored in plaintext.

# PART II

## Platform Overview

# 4. Solution Overview

Mimaric addresses the challenges outlined above through 16 integrated modules organized around the complete property lifecycle. The platform is designed as a system of record — a single source of truth that connects every stakeholder, process, and data point from initial land identification to ongoing facility operations.

## Lifecycle Stages

**1. Land Acquisition —** Pipeline management with interactive map-based location selection, suitability scoring, 5-category due diligence checklists, spatial constraint analysis (10 types), weighted feasibility assessment, and formal decision gate approvals.

**2. Planning OS —** GIS-integrated subdivision planning workspace with multiple scenario comparison, 12-type compliance rule checking, financial feasibility modeling, threaded comments, and a "Promote to Project" workflow that converts approved plans into live projects.

**3. Project Management —** 14-stage lifecycle stepper organized across 5 phase groups (Land, Design, Authority, Off-Plan, Execution). Includes building/tower management, construction site logs, document library with version control, and P&L financials per project.

**4. Sales & CRM —** Customer pipeline management with Kanban and list views, reservation workflow with expiry management, and full contract lifecycle (Draft → Sent → Signed → Cancel/Void) supporting both sale and lease contract types.

**5. Leasing & Rentals —** Ejar-compliant lease contract creation wizard, auto-generated installment schedules based on payment frequency, rent collection tracking with overdue alerts, and linked contract management.

**6. Finance —** Revenue KPI dashboard, VAT calculation (ZATCA-compliant), profit and loss analysis per unit and per project, rent collection analytics, and maintenance cost trending.

**7. Maintenance CMMS —** Work order management with Kanban and table views across 10 categories, 4 priority levels, and 6 status stages. Includes preventive maintenance scheduling with 7 recurrence types, technician assignment, SLA tracking, and cost analysis.

**8. Administration —** Multi-tenant team management with role-based access, token-based invitations, CR-based organization discovery, audit log viewer, help center (58 FAQs + 25 guides), support ticket system, and SaaS billing with 3-tier subscription plans.

# 5. Platform Module Map

The following table details all 16 platform modules with their key capabilities:

| Module | Description | Key Capabilities |
|---|---|---|
| Dashboard | Real-time analytics hub | KPI cards, occupancy donut chart, revenue trends, land pipeline chart, maintenance cost trends |
| Land Acquisition | Pre-development pipeline | Map picker, due diligence (5 categories), constraints (10 types), feasibility scoring (5-factor weighted), decision gates |
| Planning OS | GIS subdivision workspace | Spatial layers (8 types), scenarios with comparison, compliance rules (12 types), feasibility assumptions, threaded comments |
| Projects | Development management | 14-stage stepper, buildings, phases, site logs, P&L financials, document uploads |
| Unit Matrix | Inventory management | 6 unit types, bulk editing, status tracking, financial summary, contract linking |
| Customer CRM | Sales pipeline | Kanban/list views, PII encryption, pipeline stages (New→Converted), agent assignment |
| Reservations | Temporary holds | Expiry management, customer-unit linking, status workflow |
| Contracts | Agreement lifecycle | Sale + lease types, Ejar compliance, Wafi compliance, auto-numbering, escrow integration |
| Rentals | Tenancy management | Lease wizard, installment generation, payment tracking, overdue alerts |
| Finance | Financial intelligence | Revenue KPIs, VAT calculation, rent collection, P&L per unit/project |
| Maintenance | CMMS work orders | 10 categories, 4 priorities, 6 statuses, technician assignment, cost tracking |
| Preventive Maint. | Scheduled maintenance | 7 recurrence types, auto work-order generation, category tagging |
| Documents | File management | 10 categories, version history, multi-format support (PDF, images, XLSX, DOCX) |
| Help Center | Support hub | 58 FAQs (8 categories), 25 step-by-step guides, support tickets, permission requests |
| Billing | SaaS commerce | 3-tier plans, monthly/annual billing, coupons, invoices with VAT, payment methods |
| Settings | Organization config | Profile (MOC-aligned), team management, security, audit logs |

# 6. User Roles & Permissions

Mimaric implements 13 user roles organized in three tiers, with 40+ granular permissions across 15+ resource categories.

### System Tier

- **SYSTEM_ADMIN** — Full platform access including all organizations, system configuration, and platform admin panel
- **SYSTEM_SUPPORT** — Platform operations, support ticket management, and organization-level access

### Management Tier

- **COMPANY_ADMIN** — Full organization control: team, projects, finance, PII access, audit logs
- **PROJECT_MANAGER** — Projects, land pipeline, off-plan stages, maintenance, buildings
- **SALES_MANAGER** — Customer CRM with PII, contracts, reservations, inventory, pricing, launch waves
- **PROPERTY_MANAGER** — Units, leases, maintenance, preventive maintenance, tenant management
- **FINANCE_OFFICER** — Financial data, reports with export, contracts (read), pricing, feasibility

### Operational Tier

SALES_AGENT, TECHNICIAN, ENGINEERING_CONSULTANT, BUYER, TENANT, USER — each with precisely scoped permissions for their operational needs.

### Permission Model Features

- Progressive access: agents create, managers approve, admins delete
- PII access segregation: only authorized roles see decrypted personal data
- Role-filtered sidebar navigation: users see only modules they can access
- Server-side enforcement: every Server Action validates permissions before execution

# PART III

## Technical Architecture

## 7. Architecture Overview

Mimaric is built as a Turborepo monorepo with shared packages, enabling code reuse across applications while maintaining clear boundaries between concerns. The architecture prioritizes type safety, server-side security, and developer productivity.

### Monorepo Structure

The codebase is organized into two applications and four shared packages:

| Package | Name | Purpose |
|---|---|---|
| apps/web | Main Dashboard | Next.js 16 management application with 16 modules, 61 server actions, and full dashboard |
| apps/portal | Tenant/Buyer Portal | Lightweight portal for tenants to view leases and buyers to track purchases |
| packages/db | @repo/db | Prisma ORM with 80+ models, 2,500-line schema, and seed data |
| packages/ui | @repo/ui | Shared component library: Button, Input, Badge, SARAmount, RiyalIcon |
| packages/types | @repo/types | Shared TypeScript type definitions |

### Key Architectural Decisions

**Server Actions Pattern (Not REST) —** All data mutations use Next.js Server Actions with “use server” directives. This eliminates the API layer, provides automatic type safety between client and server, and ensures all business logic runs on the server with permission checks.

**Edge-Safe Auth Split —** Authentication is split into two files: auth.config.ts (edge-compatible for middleware) and auth.ts (Node.js for Prisma and bcrypt). This allows route protection to run on edge while keeping heavy operations server-side.

**Multi-Tenant Data Isolation —** Every authenticated request carries an organizationId in the JWT token. All database queries automatically filter by this ID, ensuring zero cross-tenant data leakage by design.

**Decimal Serialization —** Financial amounts use Prisma's Decimal type for precision. All server action returns are wrapped with JSON.parse(JSON.stringify()) to handle Decimal serialization across the server-client boundary.

# 8. Technology Stack

Every technology choice in Mimaric's stack is driven by specific requirements: Saudi market compliance, RTL support, type safety, and developer productivity.

| Layer | Technology | Version | Rationale |
|---|---|---|---|
| Monorepo | Turborepo | 2.8.14 | Incremental builds, shared packages, remote caching |
| Framework | Next.js | 16.1.5 | Server Components, Server Actions, edge middleware, App Router |
| UI Library | React | 19.2.0 | Concurrent features, Server Components support |
| Language | TypeScript | 5.9.2 | End-to-end type safety across full stack |
| Styling | Tailwind CSS | 4.2.1 | Utility-first CSS with RTL plugin and design token system |
| RTL | tailwindcss-rtl | 0.9.0 | Automatic logical property flipping for Arabic layout |
| Database | PostgreSQL | Supabase | ACID transactions, JSON columns, connection pooling |
| ORM | Prisma | 7.4.2 | Type-safe queries, schema-as-code, adapter pattern |
| Auth | NextAuth v5 | beta.30 | JWT strategy, edge-safe split, Credentials provider |
| Maps | Leaflet + React Leaflet | 1.9.4 | Open-source interactive mapping with drawing tools |
| Spatial | Turf.js | — | Area calculation, centroid, intersection, buffer analysis |
| Charts | Recharts | 3.8.0 | React-native charting with Arabic text support |
| Icons | Phosphor Icons | 2.1.10 | Consistent icon system with RTL mirroring rules |
| Uploads | Uploadthing | 7.7.4 | Type-safe file uploads (PDF, images, documents, up to 32MB) |
| Encryption | Node.js crypto | built-in | AES-256-GCM for PII encryption at rest |

# 9. Data Architecture

Mimaric's database comprises 80+ Prisma models organized into 10 bounded contexts, totaling over 2,500 lines of schema definition. The schema is designed with domain-driven principles, Saudi government field alignment, and financial precision.

| Bounded Context | Models | Key Models |
|---|---|---|
| Identity & Access | 8 | Organization, User, Invitation, JoinRequest, PermissionRequest, Subscription |
| Property & Development | 12 | Project, Building, Unit, Phase, DueDiligence, SiteLog, ConstraintRecord, FeasibilityAssessment, DecisionGate, ConceptPlan |
| Subdivision & Planning | 11 | SubdivisionPlan, Plot, Block, Road, UtilityCorridor, PlanningWorkspace, PlanningScenario, SpatialLayer, RuleTemplate |
| Sales & Contracts | 5 | Customer, Reservation, Contract, Lease, RentInstallment |
| Wafi Compliance | 8 | EscrowAccount, EscrowTransaction, ConstructionMilestone, MilestoneBillingItem, WafiLicense, EtmamRegistration |
| Maintenance | 2 | MaintenanceRequest, PreventiveMaintenancePlan |
| Authority & Approvals | 4 | ApprovalSubmission, ApprovalComment, ApprovalCondition, InfrastructureReadiness |
| Commercialization | 6 | InventoryItem, PricingRule, LaunchWave, Plan, Entitlement, Coupon |
| Documents & Audit | 4 | Document, DocumentVersion, AuditLog, Notification |
| Billing & Payments | 5 | Invoice, PaymentTransaction, PaymentMethod, GatewayConfig, WebhookEvent |

### Schema Design Principles

1. Saudi government field alignment: data models mirror Absher, MOC, Balady, Ejar, and Wafi field structures
2. PII encryption at field level with SHA-256 hash columns for search without decryption
3. JSON fields for flexible structured data: addresses, contacts, findings, conditions
4. Decimal type for all financial amounts ensuring SAR precision across calculations
5. Enum types for controlled vocabularies: 21 project statuses, 13 user roles, 10 maintenance categories

# 10. Multi-Tenancy Model

Mimaric implements organization-scoped data isolation through JWT claims with server-side enforcement. This ensures zero cross-tenant data leakage by design.

### Request Flow

1. JWT token carries organizationId, role, and userId claims from authentication
2. Server Action calls requirePermission("resource:action") to validate authorization
3. All Prisma queries automatically include WHERE organizationId = session.organizationId
4. Results are serialized and returned — no data from other organizations is ever accessible

### Onboarding Flows

Company registration creates a new organization with the registrant as COMPANY_ADMIN. Individual registration creates a personal workspace. Users can discover and request to join existing organizations via Commercial Registration (CR) number lookup. Admins can send token-based email invitations with pre-assigned roles.

# 11. Authentication & Authorization

## Edge-Safe Split Architecture

Authentication is split into two configurations to support both edge middleware and server-side operations:

**auth.config.ts (Edge-compatible) —** JWT callbacks, route authorization matrix, session strategy definition. Runs on edge for fast route protection without cold starts.

**auth.ts (Node.js) —** PrismaAdapter for database integration, Credentials provider with bcrypt password verification, full user lookup and role mapping.

**middleware.ts —** Route protection using the edge-safe config. Redirects unauthenticated users to login, enforces onboarding completion, and handles role-based route access.

## Authentication Flow

1. Credentials provider: email + bcrypt (cost factor 12) password verification
2. JWT strategy: role, organizationId, and onboardingCompleted stored in token claims
3. Progressive rate limiting: 5 attempts → 30s cooldown, 10 → 5 minutes, 20 → 15 minutes
4. Audit logging: LOGIN action recorded on every successful authentication with IP address
5. 40+ granular permissions across 15+ resource categories with server-side enforcement

# 12. Security & Data Protection

## Encryption at Rest

Customer PII fields — national ID, phone number, email address — are encrypted with AES-256-GCM before storage. Each encryption operation uses a unique initialization vector (IV). SHA-256 hash columns enable exact-match search on encrypted fields without exposing plaintext data. The encryption key is managed via the PII_ENCRYPTION_KEY environment variable.

## PII Masking

Server-side masking ensures unauthorized roles never receive unmasked PII. Data is masked before it leaves the server: national IDs display as “******6789”, emails as “u***@example.com”. Authorized roles can toggle visibility with a Show/Hide PII control in the UI.

## PDPL Article 32 Compliance

Every sensitive operation is logged to the AuditLog table with 8 action types: LOGIN, READ_PII, EXPORT, CREATE, UPDATE, DELETE, STATUS_CHANGE, and SETTINGS_CHANGE. Each log entry records: userId, role, resource type, resource ID, metadata (JSON), IP address, and timestamp. A dedicated audit log viewer is available to administrators. PII reads and data exports are tracked separately per PDPL Article 32 requirements.

## Password Security (NIST SP 800-63B)

Password policy follows NIST SP 800-63B guidelines: minimum 10 characters with no arbitrary complexity rules, common password blocklist checking, contextual rejection (username and email cannot appear in password), bcrypt cost factor 12 for hashing, and real-time bilingual strength feedback in the UI.

# 13. Server Actions Pattern

All data mutations in Mimaric use Next.js Server Actions — there are no REST API endpoints except for NextAuth authentication and Uploadthing file uploads. This pattern provides automatic type safety, eliminates the API surface, and ensures all business logic executes on the server.

**Standard Pattern**

Every Server Action follows a consistent pattern: declare “use server”, call requirePermission() to validate authorization, extract organizationId from the session for multi-tenant scoping, execute Prisma queries with type-safe inputs, wrap results with JSON.parse(JSON.stringify()) for Decimal serialization, and return typed results.

The platform implements 61 Server Action files covering all operations: authentication, onboarding, customer CRM, project management, unit inventory, contract lifecycle, lease management, reservations, installments, finance, maintenance, preventive maintenance, documents, land acquisition, sales analytics, search, reports, audit, notifications, organization, team management, invitations, permission requests, join requests, support tickets, user preferences, site logs, constraints, feasibility, decision gates, concept plans, subdivision, approvals, infrastructure, inventory, pricing, launch waves, planning workspaces, planning scenarios, inventory handoff, plot conversion, post-handover maintenance, Wafi consultants, milestones, escrow, Wafi contracts, Etmam sync, billing, coupons, compliance, feasibility assumptions, planning comments, planning conversion, and planning reports.

# 14. GIS & Spatial Engine

## Current Implementation

Mimaric integrates Leaflet with React Leaflet for interactive mapping, Leaflet Geoman for drawing tools and spatial editing, Turf.js for geometric analysis (area calculation, centroid, intersection, buffer), and a shapefile parser for importing external spatial data. All geometry is stored as GeoJSON in PostgreSQL JSON columns.

## Planning OS Spatial Features

The Planning OS module provides comprehensive spatial capabilities across 8 layer types: boundary, parcels, roads, zones, utilities, imported data, reference layers, and topography. Plot, Block, Road, and UtilityCorridor geometry is stored as GeoJSON with full subdivision plan versioning and geometry snapshots. Compliance results are mapped to spatial features for pass/fail visualization, and feasibility metrics (total area, sellable area, FAR, density, estimated revenue) are auto-calculated from geometry.

## Spatial Roadmap

Future enhancements include PostGIS migration for native spatial queries (ST_Contains, ST_Intersects, ST_Buffer), tile server integration for large-scale mapping, interactive map-based inventory visualization, and automated spatial validation rules for compliance checking.

# PART IV

## Saudi Government Integration

## 15. Integration Architecture

Mimaric sits at the center of the Saudi government digital ecosystem, with schema-level alignment to 10 government platforms. The integration follows a three-layer approach:

**Schema Alignment —** Data models mirror the exact field structures expected by government platforms. Customer fields align with Absher, Organization fields with MOC, Project fields with Balady, Contract fields with Ejar and Wafi.

**Compliance by Design —** Business rules enforce regulatory requirements at the application level: 5% security deposit cap for Ejar, milestone-based payment scheduling for Wafi, PDPL-compliant audit trails, NCA-aligned password policies.

**API-Ready Architecture —** Server Action stubs are prepared for each government API integration, enabling live connectivity when API access is granted. The current implementation validates all data against government requirements before future submission.

**Dual Display —** All government-facing data is stored and displayed in both Arabic and English, supporting bilingual reporting and compliance documentation.

## 16. Absher / Nafath / Yaqeen — National Identity

Mimaric's Customer model aligns with the Absher national identity platform for identity verification of buyers and tenants.

### Schema Alignment (Customer Model)

| Field | Description |
|---|---|
| nationalId | Saudi ID / Iqama number — encrypted with AES-256-GCM, SHA-256 hash for search |
| personType | SAUDI_CITIZEN, RESIDENT, GCC_CITIZEN, VISITOR |
| gender, dateOfBirth | Identity attributes matching Absher profile structure |
| nationality | ISO country code |
| address (JSON) | National Address (Subul) format with region, city, district, street, postal code |
| documentInfo (JSON) | ID document details: type, number, issue date, expiry date |

All national ID data is encrypted at rest with PII access restricted to authorized roles. Every PII read is logged per PDPL Article 32.

# 17. MOC — Ministry of Commerce

The Organization model aligns with MOC (Ministry of Commerce) for commercial registration and licensing:

| Field | Description |
|---|---|
| crNumber | 10-digit Commercial Registration number |
| unifiedNumber | Unified national number (700 number) |
| vatNumber | ZATCA VAT registration number |
| entityType | LLC, JOINT_STOCK, SOLE_PROPRIETORSHIP, etc. |
| legalForm | Legal structure classification |
| registrationStatus | ACTIVE, SUSPENDED, CANCELLED |
| capitalAmountSar | Registered capital in Saudi Riyals (Decimal) |

CR-based organization discovery allows individual users to search for and request to join existing organizations using their Commercial Registration number.

# 18. Balady — Municipal Services

The Project model aligns with Balady for municipal permits, building licenses, and project registration:

| Field | Description |
|---|---|
| parcelNumber | Municipal parcel identifier |
| deedNumber | Property deed reference from Ministry of Justice |
| landUse | 8 zoning classifications: residential, commercial, industrial, agricultural, administrative, public facility, mixed use, other |
| blockNumber, plotNumber | Municipal subdivision identifiers |
| region, city, district | Geographic hierarchy matching Balady's structure |
| latitude, longitude | Geospatial coordinates for map integration |
| boundaries (JSON) | GeoJSON polygon for property boundaries |

### Authority Approval Workflow

The ApprovalSubmission model tracks submissions to Balady, MOMRA, and SEC with threaded comments (ApprovalComment), formal conditions (ApprovalCondition), and resubmission tracking after condition fulfillment.

# 19. Ejar — Rental Compliance

Mimaric implements comprehensive Ejar compliance for residential and commercial lease contract registration. The Contract and Lease models enforce Ejar regulations at the application level.

### Schema Alignment

| Field | Ejar Requirement |
|---|---|
| ejarContractId | Link to Ejar platform contract for future API integration |
| paymentFrequency | Monthly, quarterly, semi-annual, or annual — determines installment schedule |
| securityDeposit | Capped at 5% of annual rent per Ejar regulation |
| autoRenewal | Toggle with configurable notice period (60-day default) |
| maintenanceResp. | LANDLORD or TENANT assignment per lease terms |
| noticePeriod | Days before contract expiry for renewal notification |

### Compliance Features

Auto-generated installment schedules based on selected payment frequency. Overdue tracking with notification alerts. Contract number auto-generation (LEASE-2026-XXXX). Full contract lifecycle: DRAFT → SENT → SIGNED → CANCELLED/VOID. Lease creation automatically links to an Ejar-compliant contract with all required fields.

# 20. Wafi — Off-Plan Sales

Mimaric implements a full Wafi compliance module with 8 dedicated database models covering escrow management, milestone-based payments, developer licensing, and regulatory reporting.

**EscrowAccount —** Project-level escrow accounts with bank name, account number, balance tracking, and status lifecycle. Linked to projects with deposit and withdrawal transaction logging.

**EscrowTransaction —** Financial transaction audit trail with 6 types: BUYER_DEPOSIT (auto-created on contract signing), CONTRACTOR_PAYMENT, ADMIN_EXPENSE, RETENTION, REFUND, and REVERSAL (auto-created on contract void).

**ConstructionMilestone —** Dynamic milestone definitions per project with percentage-based payment scheduling. Engineering consultant certification workflow with status tracking: PENDING → SUBMITTED → CERTIFIED → REJECTED.

**WafiLicense —** REGA developer licensing with license type, number, issue/expiry dates, developer score tracking, and registration fee management.

**EtmamRegistration —** Developer registry synchronization with quarterly reporting: escrow statements, construction progress, financial position, unit sales, and developer score. Full sync audit trail with PUSH/PULL direction and success/failure tracking.

**Delay Penalties —** Automatic calculation: 2% annual for land delivery delays, fair market rent for unit delivery delays per Wafi regulations.

# 21. ZATCA & PDPL Compliance

## ZATCA (Tax Authority)

VAT number is stored on the Organization model. VAT calculation is integrated into the financial module with invoice generation including VAT line items. The platform is architecturally prepared for ZATCA Phase 2 e-invoicing integration (QR code generation, XML invoice submission, Fatoora portal connectivity).

## PDPL (Personal Data Protection Law)

Mimaric implements privacy by design with comprehensive PDPL compliance: AES-256-GCM encryption for all PII at rest, SHA-256 hashes for encrypted field search, role-based PII access segregation with server-side masking for unauthorized roles, Article 32 audit trail with 8 action types, data minimization principles ensuring only necessary PII is collected, and bilingual privacy notices for consent management.

# 22. Compliance Matrix

The following matrix maps Mimaric platform features to the Saudi government regulations they satisfy:

| Feature | Absher | MOC | Balady | Ejar | Wafi | ZATCA | PDPL | NCA |
|---|---|---|---|---|---|---|---|---|
| Customer Identity | ✓ | | | | | | ✓ | |
| Organization Registry | | ✓ | | | | ✓ | | |
| Project Registration | | | ✓ | | | | | |
| Lease Contracts | | | | ✓ | | | | |
| Sale Contracts | | | | | ✓ | | | |
| Escrow Management | | | | | ✓ | | | |
| Milestone Billing | | | | | ✓ | | | |
| Developer Registry | | | | | ✓ | | | |
| VAT Compliance | | | | | | ✓ | | |
| PII Encryption | ✓ | | | | | | ✓ | ✓ |
| Audit Trail | | | | | | | ✓ | |
| Password Policy | | | | | | | | ✓ |
| Rate Limiting | | | | | | | | ✓ |
| Authority Approvals | | | ✓ | | | | | |
| National Address | ✓ | | ✓ | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Land Use Zoning | | | ✓ | | | | | |

# PART V

## Commercialization & Deployment

## 23. SaaS Model

Mimaric operates on a three-tier SaaS subscription model designed to serve organizations from individual property owners to large-scale developers:

| Feature | Lite | Professional | Enterprise |
|---|---|---|---|
| Target | Small property owners | PM companies | Large developers |
| Projects | Limited | Expanded | Unlimited |
| Modules | Dashboard, units, leasing, maintenance | Full: CRM, contracts, finance, reports | All + Planning OS, Wafi, off-plan |
| Support | Email | Priority | Dedicated + custom integrations |
| Billing | Monthly / Annual | Monthly / Annual | Annual / Custom |

**Commerce Infrastructure**

Monthly and annual billing cycles with automatic renewal. Coupon and discount code system with redemption tracking. Invoice generation with ZATCA-compliant VAT line items. Payment method tokenization supporting Moyasar, Hyperpay, and Paytabs gateway adapters. Webhook-based payment confirmation with idempotency logging. Entitlement system with boolean feature flags and numeric usage caps.

## 24. Bilingual Design System

### Arabic-First Approach

Mimaric defaults to RTL layout with Arabic as the primary language. The language toggle switches to LTR English mode. All UI labels use a dual format pattern with both Arabic and English values. CSS logical properties (margin-inline-start, padding-inline-end) ensure layouts work correctly in both directions.

### Typography & Visual Identity

Arabic text uses IBM Plex Arabic (Google Fonts, weights 300–700). English text uses DM Sans (Google Fonts, weights 300–700). Monospace text (contract IDs, invoice numbers, IBANs) uses IBM Plex Mono. The design system implements glass morphism with backdrop blur, semi-transparent backgrounds, and navy-tinted elevation shadows. Gradient mesh backgrounds enhance hero sections, with glow accents in green and gold.

## Localization Features

Dual date display showing both Gregorian and Hijri (1447 AH) dates via the formatDualDate() utility. SAR currency formatting with the Riyal symbol (ر.س) through a dedicated SARAmount component. Phosphor Icons with defined RTL mirroring rules for directional icons. 4px base spacing scale for consistent layout rhythm.

# 25. Deployment & CI/CD

## Infrastructure

**Hosting:** Vercel (optimized for Next.js with edge functions, serverless, and static generation).

**Database:** Supabase PostgreSQL with transaction pooler for application connections and direct URL for schema management.

**File Storage:** Uploadthing with type-safe upload routes (contracts up to 16MB, blueprints and documents up to 32MB).

## CI/CD Pipeline (GitHub Actions)

6. Install dependencies (npm ci)
7. Generate Prisma client from schema
8. ESLint linting across all packages
9. TypeScript strict type checking
10. Next.js production build with Turbo caching

## Environment Management

Turbo globalEnv declares build-time variables (DATABASE_URL, AUTH_SECRET, AUTH_TRUST_HOST, AUTH_URL, PII_ENCRYPTION_KEY) ensuring they pass through to all packages. Separate database URLs for connection pooler (application) and direct connection (schema push). Schema management uses prisma db push (schema-first approach without migration files).

# 26. Scalability Roadmap

## Near-Term (2026 Q2–Q3)

Ejar API integration for live contract registration. ZATCA Phase 2 e-invoicing with QR code generation. Payment gateway integration (Mada/SADAD) for Saudi payment methods. Mobile-responsive optimization. Buyer/Tenant portal expansion with self-service features.

## Mid-Term (2026 Q4 — 2027)

PostGIS migration for native spatial queries replacing JSON-based geometry storage. Redis for distributed rate limiting and session caching across multiple instances. Real-time notifications via WebSockets for

instant alerts. Mobile native applications (React Native) for field operations. AI-assisted feasibility analysis for automated suitability scoring.

## Long-Term (2027+)

IoT integration for smart building management (HVAC, energy, access control). Predictive maintenance using machine learning on historical work order data. Government API marketplace for third-party integrations. Multi-country expansion to GCC markets (UAE, Bahrain, Kuwait, Oman, Qatar). White-label platform option for enterprise clients.

# PART VI

## Closing

## 27. Competitive Differentiation

| Dimension | Western Platforms | Generic Arabic Tools | Mimaric |
|---|---|---|---|
| Language | English-first, Arabic bolted on | Arabic UI only | Arabic-first bilingual (RTL default) |
| Saudi Regulations | Manual compliance | Partial support | Schema-level integration (10 platforms) |
| Lifecycle Coverage | Leasing + maintenance | Single module | Full lifecycle (land → maintenance) |
| Off-Plan Development | Not supported | Not supported | 12-stage pipeline with GIS |
| Data Protection | GDPR-focused | Basic / none | PDPL + NCA native (AES-256-GCM) |
| Architecture | Legacy monolith | Varies | Cloud-native (Next.js 16, React 19) |
| Pricing | Enterprise-only ($10K+/mo) | Low-cost basic | 3-tier SaaS (accessible to SMEs) |

### Competitive Moat

5. Deep Saudi regulatory knowledge embedded in 80+ data models — not a configuration layer
6. Full lifecycle coverage eliminates multi-vendor integration complexity
7. GIS-integrated Planning OS is unique in the Saudi PropTech market
8. Arabic-first UX with RTL, Hijri dates, and SAR formatting cannot be replicated by translation layers
9. Cloud-native architecture enables rapid feature delivery with zero-downtime deployments

# 28. Conclusion

Mimaric represents a new category of PropTech platform — purpose-built for the Saudi market, fully aligned with Vision 2030's digital transformation mandate, and designed to serve as the system of record for real estate operations.

By unifying the entire property lifecycle under a single, Arabic-first, regulation-native platform, Mimaric eliminates the operational fragmentation that has constrained Saudi real estate companies for decades. From raw land identification through urban planning, construction management, off-plan sales with escrow compliance, lease management with Ejar integration, ongoing facility maintenance, and financial reporting with ZATCA compliance — every process is connected, every data point is a single source of truth, and every regulatory requirement is enforced by design.

The platform's cloud-native architecture, enterprise-grade security with PDPL and NCA compliance, GIS-integrated planning capabilities, and deep government platform alignment position it to become essential infrastructure for Saudi real estate developers, property managers, and facility management firms.

As the kingdom's $1.3 trillion real estate sector undergoes its most significant transformation in history, Mimaric provides the digital foundation that Saudi property companies need to operate efficiently, comply confidently, and scale sustainably.

**Mimaric PropTech Co. | ميماريك للتقنية العقارية**
Saudi PropTech • Automation & Management

# APPENDICES

## Appendix A: Off-Plan Development Pipeline

Mimaric implements a 12-stage off-plan development pipeline, each requiring a formal DecisionGate approval before progression:

| Stage | Name | Description |
|---|---|---|
| 1 | Raw Land Intake | Initial land identification and basic data capture with map-based location |
| 2 | Spatial Screening | Constraint analysis: 10 types (zoning, topography, flood, utility, environmental, etc.) |
| 3 | Due Diligence | 5-category feasibility assessment with weighted scoring (Legal 25%, Commercial 25%, Technical 20%, Environmental 15%, Financial 15%) |
| 4 | Development Concept | Alternative concept plans with comparison and selection |
| 5 | Subdivision Design | GIS-based plot, block, road, and utility corridor design with versioning |
| 6 | Spatial Validation | Compliance checking against 12 rule types (area, frontage, FAR, setback, etc.) |
| 7 | Authority Approvals | Submission tracking to Balady, MOMRA, SEC with comments and conditions |
| 8 | Infrastructure Readiness | 10-category readiness matrix (electricity, water, sewage, telecom, roads, etc.) |
| 9 | Inventory Structuring | Auto-generate saleable inventory from plots (7 product types, 6 statuses) |
| 10 | Pricing & Packaging | 8-type pricing engine (base, corner, park-facing, main road, phase, block, product, custom) |
| 11 | Launch Readiness | Pre-launch verification and wave planning |
| 12 | Off-Plan Launch | Phased release control with 4 statuses (planned, ready, launched, closed) |

# Appendix B: Feasibility Scoring Model

The feasibility assessment uses a weighted 5-factor scoring model:

| Factor | Weight | Assessment Criteria |
|---|---|---|
| Legal | 25% | Zoning compliance, title clarity, regulatory approvals, encumbrances |
| Commercial | 25% | Market demand, pricing potential, competition analysis, absorption rate |
| Technical | 20% | Site accessibility, infrastructure availability, construction feasibility, soil conditions |
| Environmental | 15% | Environmental impact, flood risk, topography, contamination |
| Financial | 15% | ROI projections, construction costs, market absorption, financing feasibility |

Recommendation outputs: GO (proceed with development), NO_GO (abandon or defer), CONDITIONAL (proceed with specific mitigations). Each assessment captures structured findings as JSON data for detailed analysis.

# Appendix C: Contract Lifecycle State Machine

Contract lifecycle supports both sale and lease contract types with a common state machine:

**DRAFT —** Initial creation. All fields editable. No side-effects triggered. Contract can be deleted.

**SENT —** Shared with counterparty for review. Fields locked against editing. Can return to DRAFT or advance to SIGNED.

**SIGNED —** Legally binding. Triggers side-effects: for lease contracts, auto-creates Lease record with installment schedule; for sale contracts with Wafi compliance, auto-creates BUYER_DEPOSIT escrow transaction. Cannot be edited.

**CANCELLED —** Pre-signature cancellation. No side-effects to reverse. Terminal state.

**VOID —** Post-signature nullification. For Wafi contracts, auto-creates REVERSAL escrow transaction. Terminal state. Requires COMPANY_ADMIN permission.

# Appendix D: Glossary

| Term | Arabic | Description |
|---|---|---|
| Absher | أبشر | National digital identity platform (Ministry of Interior) |
| Balady | بلدي | Municipal services platform for permits and licensing |
| Ejar | إيجار | Rental contract registration platform (REGA) |
| Etmam | إتمام | Developer registry and reporting platform |
| MOC | وزارة التجارة | Ministry of Commerce — commercial registration and licensing |
| NCA | الهيئة الوطنية للأمن السيبراني | National Cybersecurity Authority — security standards |
| PDPL | نظام حماية البيانات الشخصية | Personal Data Protection Law — Saudi data privacy regulation |
| REGA | الهيئة العامة للعقار | Real Estate General Authority — sector regulator |
| Wafi | وافي | Off-plan property sales regulation platform (REGA) |
| ZATCA | هيئة الزكاة والضريبة والجمارك | Tax authority — VAT, e-invoicing, customs |
| CR | سجل تجاري | Commercial Registration — business license number |
| SAR | ريال سعودي | Saudi Riyal — national currency |
| VAT | ضريبة القيمة المضافة | Value Added Tax (15% in Saudi Arabia) |
| PII | — | Personally Identifiable Information |
| RBAC | — | Role-Based Access Control |
| CMMS | — | Computerized Maintenance Management System |
| GIS | — | Geographic Information System |
| SaaS | — | Software as a Service |

*— End of Document —*